

Design a two-stage op-amp shown below, where  $V_{DD} = 3V$ , total power consumption = 6mW, maximum output swing = 2.6V, channel length =  $0.5\mu m$ ,  $V_{THN} = 0.6V$ ,  $V_{THP} = -0.7V$ ,  $\lambda_n = 0.1V^{-1}$ ,  $\lambda_p = 0.2V^{-1}$ ,  $k'_n = 96\mu A/V^2$ ,  $k'_p = 32\mu A/V^2$ . Ignore channel modulation effect for DC calculation and body effect.

Fig. 1

- 1) Assume 1mA current is allocated to the second stage and equal overdrive voltage for M5 and M6, determine the aspect ratios (W/L) of these two transistors;
- 2) Calculate small-signal gain of the second stage;
- 3) Assuming the remaining 1mA flows through M7, find the aspect ratios (W/L) of M3 and M4 so that  $V_{GS3}=V_{GS5}$  to minimize the systematic offset.
- 4) Calculate the aspect ratios (W/L) of M1 and M2 such that the overall voltage gain of the op-amp is 1000.
- 5) Find the pole and the zero frequencies, and obtain the transfer function by considering only  $C_y$ ,  $C_{gd5}$ , and  $C_L$ .

Fig. 1

Design a two-stage op-amp shown below, where  $V_{DD} = 3V$ , total power consumption = 6mW, maximum output swing = 2.6V, channel length = 0.5 $\mu m$ ,  $V_{THN} = 0.6V$ ,  $V_{THP} = -0.7V$ ,  $\lambda_n = 0.1V^{-1}$ ,  $\lambda_p = 0.2V^{-1}$ ,  $k'_n = 96\mu A/V^2$ ,  $k'_p = 32\mu A/V^2$ . Ignore channel modulation effect for DC calculation and body effect.

Solution:

- 1) Assume 1mA current is allocated to the second stage and equal overdrive voltage for M5 and M6, determine the aspect ratios (W/L) of these two transistors;

Maximum output swing = 2.6V & equal  $V_{OD}$  for M5, M6:

$$I_{D,M5} = \frac{k'_p}{2} \left( \frac{W}{L} \right)_{M5} V_{OD,M5}^2 \Rightarrow V_{OD,M5} = 0.2V \Rightarrow \left( \frac{W}{L} \right)_{M5} = 1306$$

$$I_{D,M6} = \frac{k'_n}{2} \left( \frac{W}{L} \right)_{M6} V_{OD,M6}^2 \Rightarrow V_{OD,M6} = 0.2V \Rightarrow \left( \frac{W}{L} \right)_{M6} = 373$$

Design a two-stage op-amp shown below, where  $V_{DD} = 3V$ , total power consumption = 6mW, maximum output swing = 2.6V, channel length = 0.5 $\mu m$ ,  $V_{THN} = 0.6V$ ,  $V_{THP} = -0.7V$ ,  $\lambda_n = 0.1V^{-1}$ ,  $\lambda_p = 0.2V^{-1}$ ,  $k'_n = 96\mu A/V^2$ ,  $k'_p = 32\mu A/V^2$ . Ignore channel modulation effect for DC calculation and body effect.

Solution:

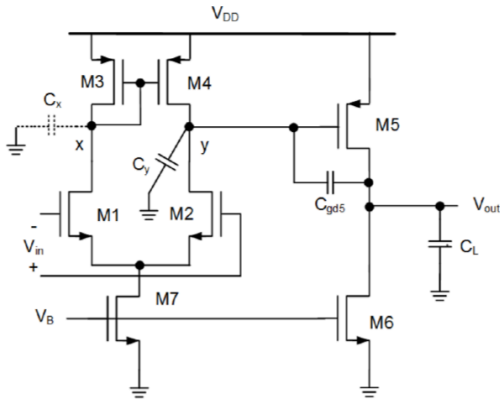
- 2) Calculate small-signal gain of the second stage;

$$g_{m,M5} = \sqrt{2k'_p \left(\frac{W}{L}\right)_{M5} I_{D,M5}} = 10\text{mA/V}$$

$$r_{o,M5} = \frac{1}{\lambda_p \times I_{D,M5}} = 5k\Omega, r_{o,M6} = \frac{1}{\lambda_n \times I_{D,M6}} = 10k\Omega$$

$$A_{V2} = g_{m,M5} \times (r_{o,M5} // r_{o,M6}) = 33.33$$

Fig. 1



\* $C_y = 0.1\text{pF}$ ,  $C_{gd5} = 0.05\text{pF}$ ,  $C_L = 0.3\text{pF}$

Fig. 1

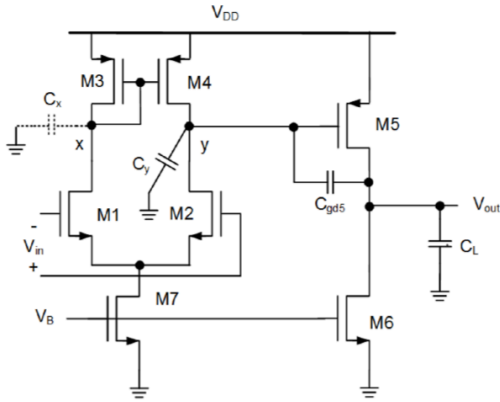
Design a two-stage op-amp shown below, where  $V_{DD} = 3\text{V}$ , total power consumption =  $6\text{mW}$ , maximum output swing =  $2.6\text{V}$ , channel length =  $0.5\mu\text{m}$ ,  $V_{THN} = 0.6\text{V}$ ,  $V_{THP} = -0.7\text{V}$ ,  $\lambda_n = 0.1\text{V}^{-1}$ ,  $\lambda_p = 0.2\text{V}^{-1}$ ,  $k'_n = 96\mu\text{A/V}^2$ ,  $k'_p = 32\mu\text{A/V}^2$ . Ignore channel modulation effect for DC calculation and body effect.

Solution:

3) Assuming the remaining  $1\text{mA}$  flows through  $M_7$ , find the aspect ratios ( $W/L$ ) of  $M_3$  and  $M_4$  so that  $V_{GS3} = V_{GS5}$  to minimize the systematic offset.

To minimize offset,  $M_3$  should be mirroring  $M_5$  based on the current mirror ratio. Hence,  $M_3$  should be half of  $M_5$  aspect ratio.

$$\left(\frac{W}{L}\right)_{M3} = 0.5 \left(\frac{W}{L}\right)_{M5} = 653$$



\* $C_y = 0.1\text{pF}$ ,  $C_{gd5} = 0.05\text{pF}$ ,  $C_L = 0.3\text{pF}$

Fig. 1

Design a two-stage op-amp shown below, where  $V_{DD} = 3\text{V}$ , total power consumption =  $6\text{mW}$ , maximum output swing =  $2.6\text{V}$ , channel length =  $0.5\mu\text{m}$ ,  $V_{THN} = 0.6\text{V}$ ,  $V_{THP} = -0.7\text{V}$ ,  $\lambda_n = 0.1\text{V}^{-1}$ ,  $\lambda_p = 0.2\text{V}^{-1}$ ,  $k'_n = 96\mu\text{A/V}^2$ ,  $k'_p = 32\mu\text{A/V}^2$ . Ignore channel modulation effect for DC calculation and body effect.

Solution:

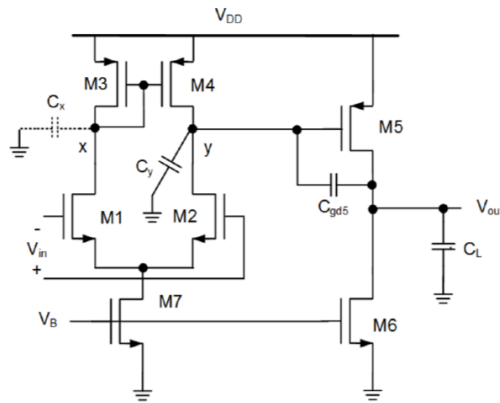
4) Calculate the aspect ratios ( $W/L$ ) of  $M_1$  and  $M_2$  such that the overall voltage gain of the op-amp is 1000.

Given  $A_{V2} = 33.33$  and total gain is 1000.  $A_{V1} = 30$ .

$$r_{o,M4} = \frac{1}{\lambda_p \times I_{D,M3}} = 10\text{k}\Omega, r_{o,M2} = \frac{1}{\lambda_n \times I_{D,M1}} = 20\text{k}\Omega$$

$$A_{V1} = g_{m,M1} \times (r_{o,M4} // r_{o,M2}) = 30 \Rightarrow g_{m,M1} = 4.5\text{mA/V}$$

$$g_{m,M1} = \sqrt{2k'_n \left(\frac{W}{L}\right)_{M1} I_{D,M1}} = 4.5\text{mA/V} \Rightarrow \left(\frac{W}{L}\right)_{M1} = 151$$



\* $C_y = 0.1\text{pF}$ ,  $C_{gd5} = 0.05\text{pF}$ ,  $C_L = 0.3\text{pF}$

Fig. 1

Design a two-stage op-amp shown below, where  $V_{DD} = 3\text{V}$ , total power consumption =  $6\text{mW}$ , maximum output swing =  $2.6\text{V}$ , channel length =  $0.5\mu\text{m}$ ,  $V_{THN} = 0.6\text{V}$ ,  $V_{THP} = -0.7\text{V}$ ,  $\lambda_n = 0.1\text{V}^{-1}$ ,  $\lambda_p = 0.2\text{V}^{-1}$ ,  $k'_n = 96\mu\text{A/V}^2$ ,  $k'_p = 32\mu\text{A/V}^2$ . Ignore channel modulation effect for DC calculation and body effect.

Solution:

5) Find the pole and the zero frequencies, and obtain the transfer function by considering only  $C_y$ ,  $C_{gd5}$ , and  $C_L$ .

$$\frac{v_{out}(s)}{v_{in}(s)} = \frac{g_{m2} R_1 g_{m5} R_2 (1 - s/\omega_z)}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})} \quad \omega_{p1} = \frac{1}{R_1 C_1}, \quad \omega_{p2} = \frac{1}{R_2 C_2} \quad \text{and} \quad \omega_z = \frac{g_{m5}}{C_{gd5}}$$

$$R_1 = (r_{o2} // r_{o4}) = (20\text{k} // 10\text{k}) = 6.666\text{k}$$

$$R_2 = (r_{o5} // r_{o6}) = (5\text{k} // 10\text{k}) = 3.333\text{k}$$

$$C_1 = C_y + (1 + A_{V2})C_{gd5} = 0.1\text{pF} + (1 + 33.333) \times 0.05\text{pF} = 1.816\text{pF}$$

$$C_2 = C_L + C_{gd5} = 0.3\text{pF} + 0.05\text{pF} = 0.35\text{pF}$$

$$\omega_{p1} = 82.6\text{Mrad/s} \quad (13.14\text{MHz}), \quad \omega_{p2} = 857.2\text{Mrad/s} \quad (136.4\text{MHz})$$

$$\omega_z = 200\text{Grad/s} \quad (31.8\text{GHz})$$

## Question 2

For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient,  $\beta$ , is independent of frequency and the open-loop transfer function of the operational amplifier is

$$A(s) = \frac{A_{DC}}{1 + s/\omega_{3dB}}$$

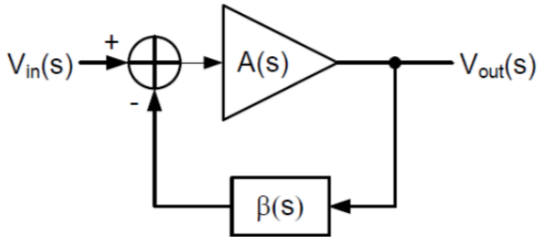


Fig. 2

- 1) Find the unity-gain bandwidth (bandwidth at gain = 1) of the op-amp
- 2) Derive the closed-loop transfer function,  $A_{cl}(s) = V_{out}(s)/V_{in}(s)$ .
- 3) Find the 3dB bandwidth of the feedback amplifier and comment on the result.
- 4) Show that the gain-bandwidth product of the closed-loop amplifier is a constant and equals its unity-gain bandwidth of the op-amp.

For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient,  $\beta$ , is independent of frequency and the open-loop transfer function of the operational amplifier is

$$A(s) = \frac{A_{DC}}{1 + s/\omega_{3dB}}$$

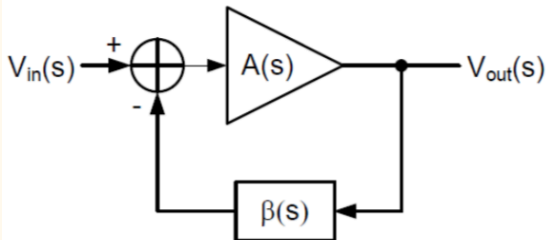


Fig. 2

Solution:

- 1) Find the unity-gain bandwidth (bandwidth at gain = 1) of the op-amp

$$\left. \frac{A_{DC}}{(1 + j\omega_{unity}/\omega_{3dB})} \right|_{\text{large } \omega_{unity}} \approx \left| \frac{A_{DC}}{(j\omega_{unity}/\omega_{3dB})} \right| = 1 \Rightarrow \omega_{unity} = A_{DC} \times \omega_{3dB}$$

For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient,  $\beta$ , is independent of frequency and the open-loop transfer function of the operational amplifier is

$$A(s) = \frac{A_{DC}}{1 + s/\omega_{3dB}}$$

Solution:

- 2) Derive the closed-loop transfer function,  $A_{cl}(s) = V_{out}(s)/V_{in}(s)$ .

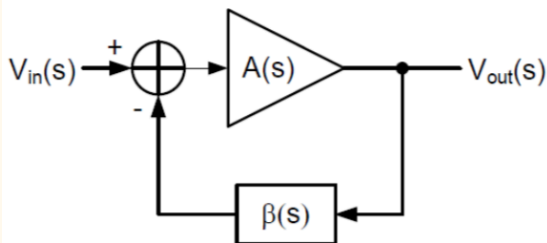


Fig. 2

$$A_{cl}(s) = \frac{A(s)}{1 + \beta A(s)} = \frac{\frac{A_{DC}}{(1 + s/\omega_{3dB})}}{1 + \frac{\beta A_{DC}}{(1 + s/\omega_{3dB})}} = \frac{A_{DC}}{1 + s/\omega_{3dB} + \beta A_{DC}} = \frac{\frac{A_{DC}}{(1 + \beta A_{DC})}}{1 + s/[\omega_{3dB}(1 + \beta A_{DC})]}$$

For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient,  $\beta$ , is independent of frequency and the open-loop transfer function of the operational amplifier is

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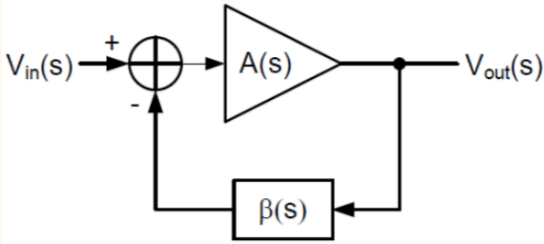


Fig. 2

Solution:

3) Find the 3dB bandwidth of the feedback amplifier and comment on the result.

$$A_{cl}(s) = \frac{A(s)}{1 + \beta A(s)} = \frac{\frac{A_{DC}}{(1 + s/\omega_{3dB})}}{1 + \frac{\beta A_{DC}}{(1 + s/\omega_{3dB})}} = \frac{A_{DC}}{1 + s/\omega_{3dB} + \beta A_{DC}} = \frac{\frac{A_{DC}}{(1 + \beta A_{DC})}}{1 + s/[\omega_{3dB}(1 + \beta A_{DC})]}$$

The closed-loop 3dB bandwidth is  $\omega_{3dB}(1 + \beta A_{DC})$ . Through feedback, we widen the bandwidth by loop gain factor of  $(1 + \beta A_{DC})$ .

For the feedback amplifier shown in Fig. 2 below, assume that the feedback coefficient,  $\beta$ , is independent of frequency and the open-loop transfer function of the operational amplifier is

$$A(s) = \frac{A_{DC}}{1 + s/\omega_{3dB}}$$

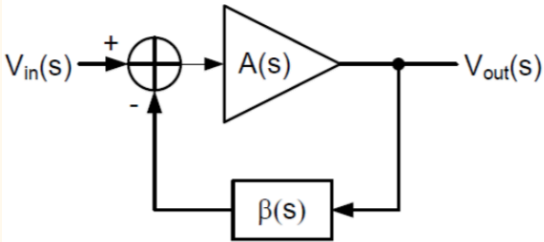


Fig. 2

Solution:

4) Show that the gain-bandwidth product of the closed-loop amplifier is a constant and equals its unity-gain bandwidth of the op-amp.

The gain is  $\frac{A_{DC}}{(1 + \beta A_{DC})}$ , the bandwidth is  $\omega_{3dB}(1 + \beta A_{DC})$ . The gain-bandwidth product became  $A_{DC} \times \omega_{3dB}$ , which is a constant and equals to unity gain bandwidth of the opamp.